CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Mechanical Measurement & Metrology (ME 345)

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Mechanical Measurement & Metrology (ME 345) **Date:**

EXPERIMENT NO. 1
PRECISION MEASURING INSTRUMENTS

AIM:
Review of precision measuring instruments.

OBJECTIVES:
1. To know the technical specifications, working principles and applications of PMI.
2. To understand the standards of measurements.
3. To study errors in measurements.

APPARATUS:
All the instruments available in the lab.

THEORY:
Metrology is a science of measurement. Engineering metrology is a part of Metrology. Engineering metrology deals with the length and angle measurement. Legal metrology deals with the legal and statutory requirements of the metrology. There are primary, secondary tertiary and working standards in the world. Instruments have to be calibrated at specific times for the satisfactory working of the instrument. Instrument should have sufficient accuracy and precision.

QUESTIONS:
1. List the objective of Metrology.
2. Explain Generalized Measurement System and its Elements.
3. What is standard? Explain the various standards of measurement based upon accuracy.
4. Differentiate between line standard and end standard.
5. Write a note on following points:
   1) Accuracy 2) Precision. 3) Repeatability 4) Threshold

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EXPERIMENT NO. 2
LINEAR MEASUREMENT

AIM:
Linear measurement with different linear measuring instruments.

OBJECTIVES:
1. To know the working principles and applications of LMI.
2. To understand how to eliminate errors in linear measurements.
3. To study method of selection of LMI.

APPARATUS:
1. Micrometer
2. Vernier caliper
3. Vernier height gauge
4. Vernier depth gauge
5. Telescopic gauge
6. Bore gauge

THEORY:
Vernier caliper works on the principle of minor difference in the two scales i.e. main scale and the vernier scale. Micrometer is operating on the principle of screw and nut. Vernier depth gauge measures the depth and the Vernier height gauge measures the height of the component. Outside micrometer is used to measure the outer dimension and the inside micrometer to measure the inside dimension. Bore and telescopic gauge measure the inner cavity. These two are the indirect measuring instruments. Micrometer and vernier caliper are the end standards.

OBSERVATION TABLE:

Instrument: Vernier caliper

Sr. No.	Dimension (mm)	Sr. No.	Dimension (mm)	Sr. No.	Dimension (mm)
1		11		21	
2		12		22	
3		13		23	
4		14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	

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COMPUTATION TABLE:

Sr. No.	**Dimension** x_i	**Frequency** f_i	$x_i * f_i$	$\bar{x}_i$	$(x_i - \bar{x}_i)^2$
			$\sum x_i * f_i$		$\sum (x_i - \bar{x})^2$

<u>Instrument: Micrometer</u>

Sr. No.	**Dimension (mm)**	**Sr. No.**	**Dimension (mm)**	**Sr. No.**	**Dimension (mm)**
1		11		21	
2		12		22	
3		13		23	
4		14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	

Sr, No.	**Dimension** x_i	**Frequency** f_i	$x_i * f_i$	$\bar{x}_i$	$(x_i - \bar{x})^2$
			$\sum x_i * f_i$		$\sum (x_i - \bar{x})^2$

CALCULATION:

1. Std. Deviation = $\sqrt{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}}$
2. Variance = (std. deviation.) ^{2}

GRAPH:

f_i Vs x_i

CONCLUSION:

QUESTIONS:

1. List various LMIs that you have used in Lab.
2. State principles of Vernier caliper and Micrometer with net sketch.
3. List the precautions to be taken while measuring.
4. Explain in brief Vernier Depth Gauge.

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EXPERIMENT NO. 3
ANGULAR MEASUREMENT

AIM:
Angular measurements with sine bar, vernier bevel protractor.

OBJECTIVES:

1. To know the working principles and applications of sine bar.
2. To learn the uses of vernier bevel protractor.
3. To learn the use of combination set

.

APPARATUS:
1. Sine bar
2. Spirit level
3. Vernier bevel protractor
4. Slip gauges, combination set
5. Vernier height gauge

THEORY:
The Angle is defined as the opening between two lines, which meet at a point. The Vernier bevel protractor can read to the accuracy of 5’. Sine bar is use for the accurate angle measurement and to locate the work to a given angle. Sine bar is reliable for angle less than 15° and it becomes in accurate as the angle increases. It is impractical to use sine bar for angle above 45°Angle gauges are used to measure the angle to the accuracy of 3”.

OBSERVATION TABLE:
Specimen – 1

Sr. No.	Instrument	Length	Height	Taper angle (°)

CONCLUSION:

QUESTIONS:

1. List various angle-measuring instruments.
2. What are the various line standard angular measuring devices?
3. Name the parts of a universal bevel protractor and state the functions of each.
4. How will you calculate the LC of vernier bevel protractor?
5. Differentiate between angle gauges and slip gauges.
6. Draw the set up used for measurement of angles with sine bar.
7. Why is sine bar not preferred for measuring angle more than 45°?
8. Calculate the gauge block buildup required to set a 10 cm sine bar to an angle of 30°.
9. Describe the principle, working and uses of a clinometer.
10. Describe the principle and working of an autocollimator.

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EXPERIMENT NO. 4
MICROMETER CALIBRATION

AIM:
To calibrate the micrometer.

OBJECTIVE:
To check the error in reading of micrometer with the help of slips gauges.

APPARATUS:
1. Micrometer
2. Slip gauges.

THEORY:
It is the process of framing / reframing the scale of instrument by applying some standardizes signals. Instruments are calibrated at 20° centigrade. During calibration of micrometer it has to be checked for parallelism of anvils, flatness of anvils, play between screw and nut, etc.

OBSERVATION TABLE:
Micrometer range:
Least count:

<u>Table – A</u>

Sr. No.	Slip gauge (mm)	Micrometer reading (mm)		Error (mm)		Correction (mm)	
		Increasing	Decreasing	Increasing	Decreasing	Increasing	Decreasing
1.							
2.							
3.							
4.							
5.							

<u>Table – B</u>

Sr. No.	Slip gauge (mm)	Micrometer reading (mm)		Error (mm)		Correction (mm)	
		Increasing	Decreasing	Increasing	Decreasing	Increasing	Decreasing
1.							
2.							
3.							
4.							
5.							

GRAPH/S:
1. From table A: Error V/S Slip gauge value
2. From table A: Micrometer reading V/S Slip gauge value
3. From table A: Correction V/S Micrometer reading
4. From table B: Error V/S Slip gauge value.

CONCLUSION:

QUESTIONS:

1. Define calibration. Explain the need of calibration.
2. What are the probable sources of error in functioning of micrometer?
3. Describe the procedure used to calibrate the micrometer.
4. Explain in brief Micrometer with dial gauge.

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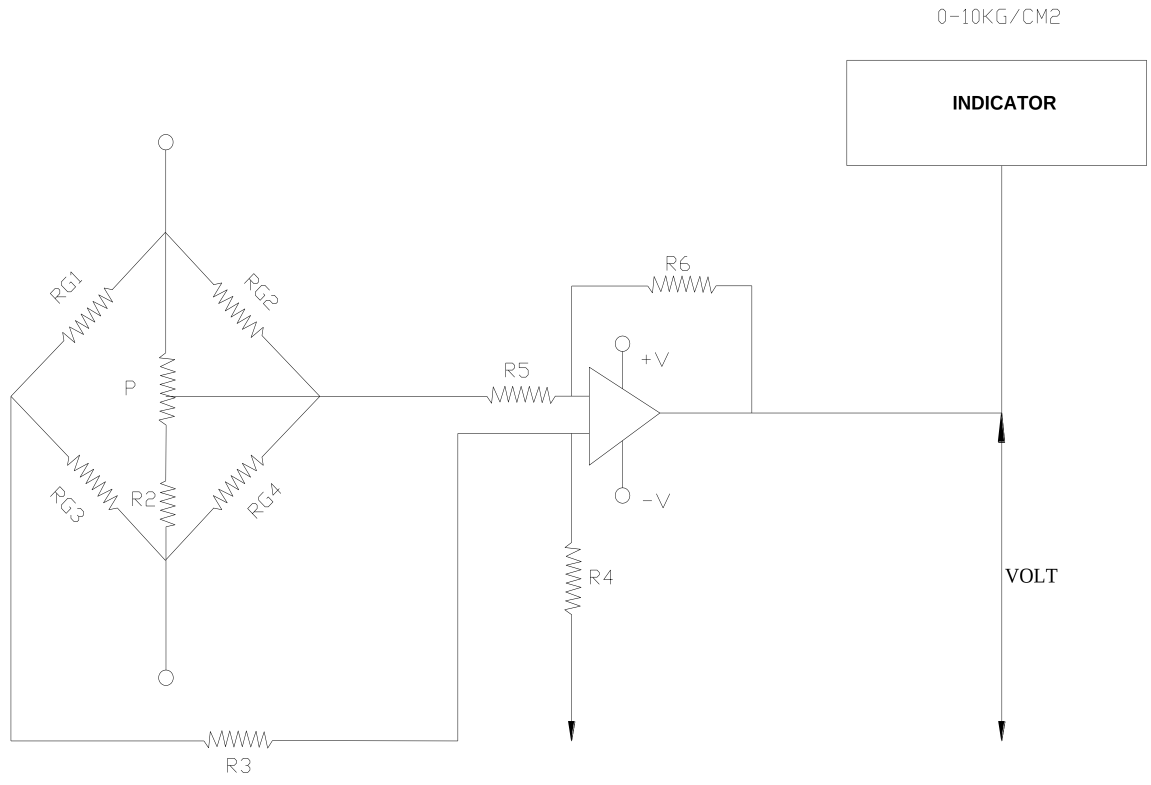
EXPERIMENT NO. 5
PRESSURE MEASUREMENT

AIM:
Pressure measurement

OBJECTIVE:
To Measure of pressure & output voltage of piezoresistive type pressure transducer on application of force.

APPARATUS:
1. Pressure cell demonstrator
2. Digital millimeters
3. Air pressure pump

CIRCUIT DIAGRAM:



THEORY:

When pressure, force or acceleration is applied to quartz crystal, a charge is developed across the crystal faces. The fundamental difference between these crystal sensors and static force devices such as strain gauges is that the electrical signal generated by crystal decays rapidly. So crystal sensors are used for dynamic measurement. i.e. to measure the pressure of blast, explosion, etc.

OBSERVATION TABLE:

Sr. No.	Pressure (Kg)	Output voltage (mV)
1.		
2.		
3.		
4.		

CONCLUSION:

QUESTIONS:

1. Why manometers are treated as standards for pressure and differential pressure measurements?
2. Explain briefly the following pressure measuring instruments
   a. Diaphragm Gauge
   b. Bellows
   c. Electromagnetic type
3. What do you mean by low pressures and vacuum?
4. Describe the construction, working and theory of McLeod gauge for measurement of vacuum. List its advantages and disadvantages.
5. How are very high pressure measured? Explain briefly with a neat sketch the construction and working of a baridgman gauge used for measuring high pressure?

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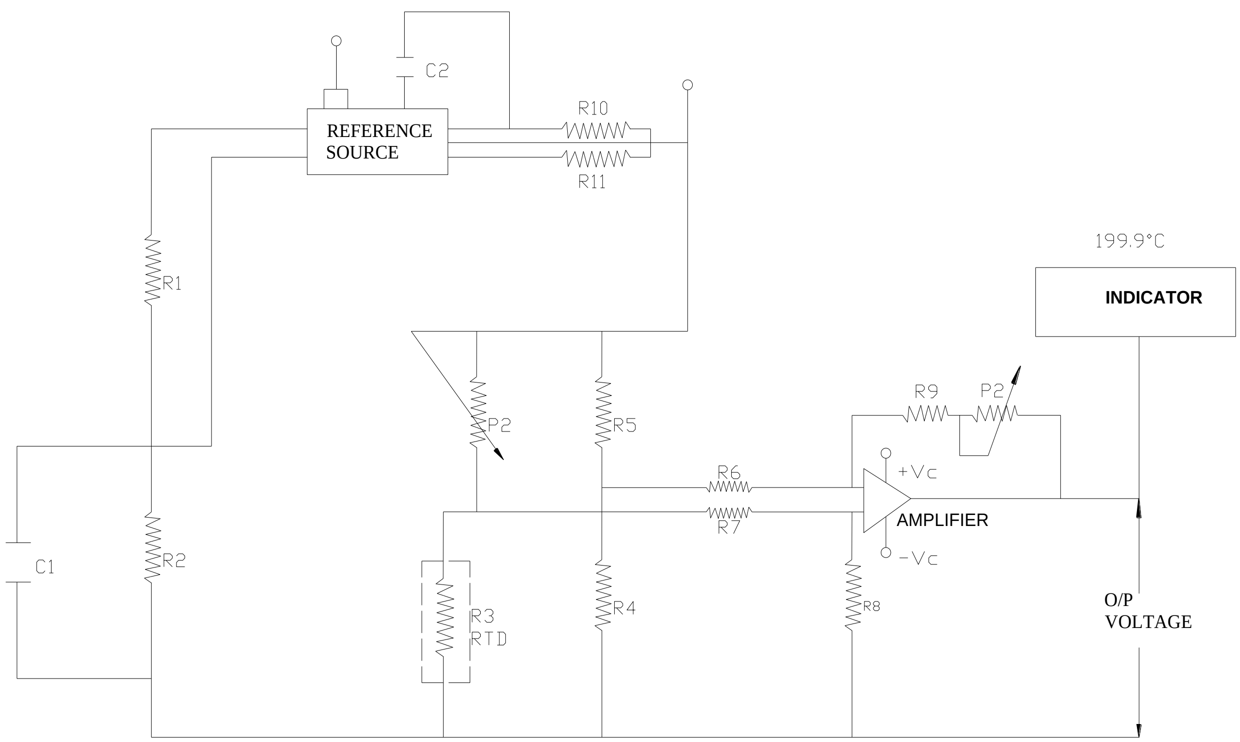
EXPERIMENT NO. 6
TEMPERATURE MEASUREMENT

AIM:
Temperature measurement

OBJECTIVE:
To determine the relationship between resistance and temperature of RTD (Resistance Temperature Detector)

APPARATUS:
1. RTD kit
2. RTD sensor
3. Electrical cattle

CIRCUIT DIAGRAM:



THEORY:

RTD exhibits the change in its resistance with the change in surrounding temperature, applying this knowledge; the temperature of a particular device or element can be measured using a meter, which is calibrated to display temperature with property signal conditioning circuitry. This further can be used with relaying or other circuit to control the specific device. RTD has a linear temperature v/s resistance characteristic & hence no linearization circuitry is required. The RTD is used for this practical is most popular & widely one i.e. PT_{100} is 100Ω at zero degree Celsius.

OBSERVATION TABLE:

Sr. No.	Temperature (°)	Resistance (kΩ)	Voltage (mV)

CONCLUSION:

QUESTIONS:

1. Classify the temperature-measuring instrument & indicate approximate temperature range of each category.
2. Enumerate the sources of error in thermocouple and explain how they are prevented?
3. Write a short note on “Resistance Temperature Detectors”.
4. Give comparison between “thermistors” and “metal resistor”.
5. Explain the factors that influence the response of a temperature-sensing device.

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Mechanical Measurement & Metrology (ME 345) **Date:**

EXPERIMENT NO. 7
SURFACE ROUGHNESS MEASUREMENT

AIM:
To study surface finish measurement with surface tester.

OBJECTIVE:
1. To know the principle elements of surface.
2. To learn different methods of surface measurement.
3. To know different surface measurement instruments.

APPARATUS:
1. Surface tester SJ201P
2. Granite surface plate
3. CI surface plate
4. Slip gauge, etc.

THEORY:
Whatever may be the manufacturing process, it is not possible to produce perfectly smooth surface. The imperfection & irregularities are bound to occur. Vibrations, work piece material, machining methods, operator, cutting tool, cutting conditions, type of machines are the variables, which affect the surface roughness.

OBSERVATION TABLE:

Sr. No.	Component name	Surface roughness Ra value (μm)					
		1	2	3	4	5	Average
1.							
2.							
3.							
4.							
5.							

CONCLUSION:

QUESTIONS:

1. Explain various elements of surface tester.
2. Explain the following:
   i. Mean line of profile
   ii. Center line of profile
   iii. Sampling length
   iv. CLA value
   v. RMS value
   vi. R_Z value
3. Describe conventional method of measuring surface finish.
4. Write short note on following:
   i. Profilometer
   ii. Tomlinson surface meter
   iii. Taylor – Hobson – Talysurf
5. State the adverse effect of poor surface finish.

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EXPERIMENT NO. 8
TO STUDY STRAIN GAUGE TRANSDUCER.

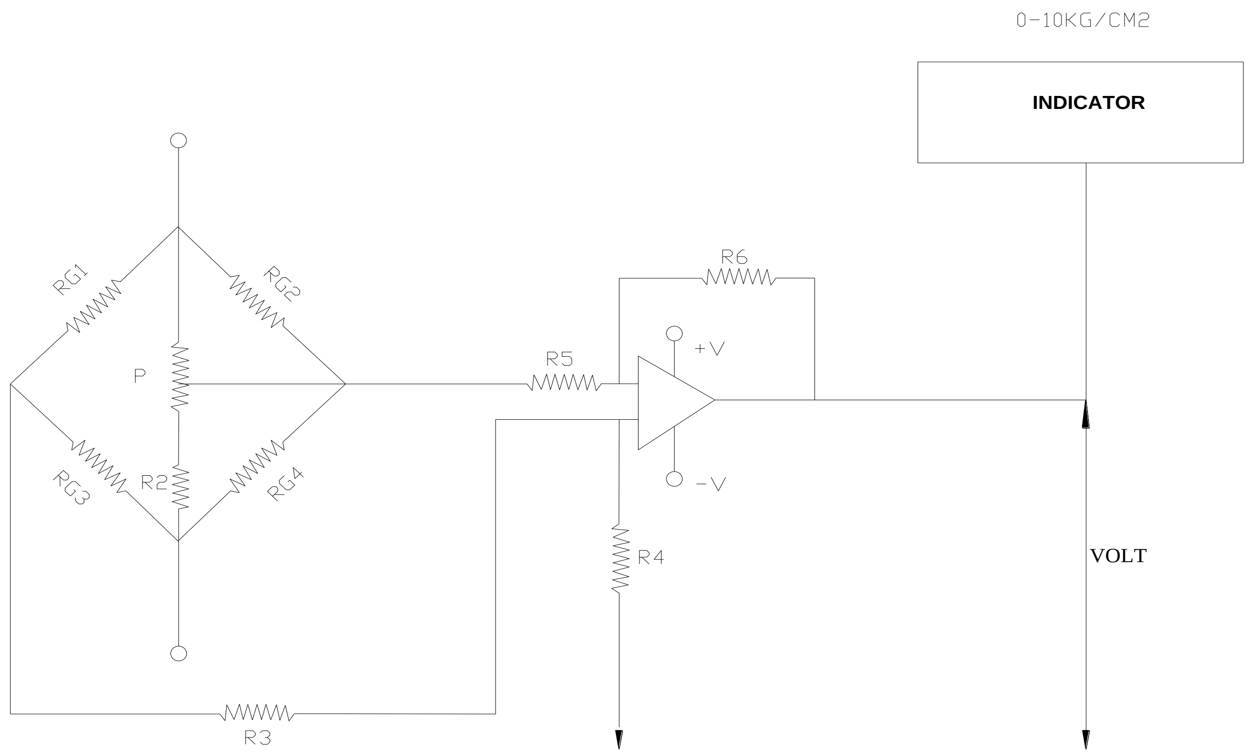
OBJECTIVE:

To see the response of strain gauge output voltage is calibrate in terms of strain on application of weight.

APPARATUS:

1. Load cell
2. Weights: 50g, 100g, 200g, and 500g, DMM.

CIRCUIT DIAGRAM:



THEORY:

If a metal conductor is stretched or compressed its resistance changes, on account of that both length and diameter of the conductor change. There is also a change in value of the resistivity of the conductor when it is strained. Therefore, a resistance strain gauge is used for the measurement of strain.

Strain=(dR/R)/(dL/L)

Where, L= length of strain gauge

dL= change in length

R=resistance of strain gauge

dR=change in resistance

Stain is usually expressed in micro strain= 1 mm per meter

OBSERVATION TABLE:

Sr. No.	Weight (gram)	Output voltage (mV)

CONCLUSION:

QUESTIONS:

1. Define strain. List some practical situation where strain measurement becomes essential?
2. List main requirements of strain gauge and mention the type of gauge, which meets most of these requirements.
3. Define the gauge factor of resistance strain gauge. And factors affecting on it.
4. Distinguish between bonded and unbounded type strain gauge.

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT NO. 9
GEAR TOOTH THICKNESS MEASUREMENT

AIM:
Measurement of gear tooth thickness with gear tooth vernier caliper.

OBJECTIVE:
1. To know the terminology of gear.
2. To learn manufacturing errors in gear element.
3. To determine actual profile of the gear tooth.

APPARATUS:
1. Gear tooth vernier caliper,
2. Gears.

THEORY:
Gear is a very important component in power or motion transmission. The transmission efficiency of the gear is about 99%. Error in the elements of gear interferes with the efficient working of operation of the equipment using them. The accuracy of gears, both as to their geometrical forms, size has a considerable effect on smoothness of working, freedom from noise and length of working life.

CALCULATION:
1. $m = d_o/(T+2)$
2. $h = m+(T*m/2)[1-\cos(90/T)]$
3. $w = Tm \sin(90/T)$

where, h = chordal depth
w = chordal width
T = number of teeth
m = module

OBSERVATION TABLE:

Table A

	Theoretical	Actual
Chordal depth		
Chordal width		

Table B

Chordal depth						
Chordal width						

CONCLUSION:

QUESTIONS:

1. List the different methods of making gears.
2. List different elements of gear in which errors may present.
3. List different methods used for gear tooth thickness measurement and explain in brief constant chord method with neat sketch.
4. List the various Gear errors.
5. Calculate the setting of a gear tooth vernier caliper for a straight spur gear having 40 teeth and module 4.

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Mechanical Measurement & Metrology (ME 345) **Date:**

EXPERIMENT NO. 10
COORDINATE MEASURING MACHINE

AIM:
To Study about Coordinate Measuring Machine.

OBJECTIVES:
1. To study construction of Coordinate Measuring Machines.
2. To understand the procedure to carry out measurement on Coordinate Measuring Machines.
3. To know the applications of Coordinate Measuring Machines.

THEORY:
A coordinate measuring machine is a device for measuring the physical geometrical characteristics of an object. This machine may be manually controlled by an operator or it may be computer controlled.
A typical CMM consists of
- Probe
- Mechanical Structure
- Control Unit
- Software

QUESTIONS:
1) Explain construction of Coordinate Measuring Machines.
2) Give Classification of Coordinate Measuring Machines in details.
3) Write-down Application of Coordinate Measuring Machines.
4) Explain procedure to carry out measurement with Coordinate Measuring Machines.

Marks obtained:	Signature of faculty:	Date: